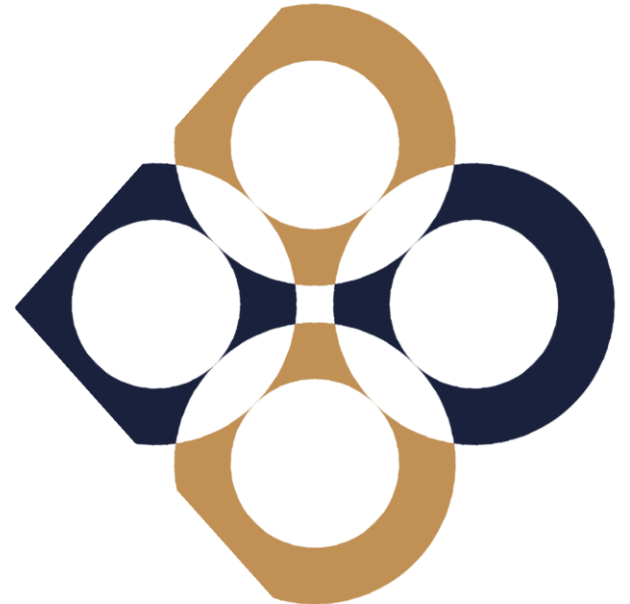


Capital Structure I.

International Corporate Finance – 4

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Based on:
Berk – DeMarzo (2017): Corporate Finance. Fourth edition, Pearson – chapter 14



Financing policy can be achieved only by a given investment policy.

Financing policy can be divided into two major parts:

- Dividend policy (Payout policy)

- Capital structure policy

Main goal of the financing policy is to find the optimal capital structure, where the value of the firm is maximal.

Optimal capital structure → value of the firm is the highest → present value of the generated cash flows are the highest

Capital structure is measured by the leverage:

$$L = \frac{D}{V} ; L = \frac{D}{E + D} ; L = \frac{D}{E} ; L = \frac{V}{E}$$

L: leverage

D: market value of debt

E: market value of equity

V: market value of the firm

Patterns of Corporate Financing

Firms may raise funds through
external sources or
plow back profits rather than distribute them to shareholders.

Should a firm elect external financing, they may choose between
debt or
equity sources.

Investment Policy

During an investment decision, the *future CF* and the *discount rate* is important.

Main problem: one cannot observe on the market the expected rate of return for the assets.

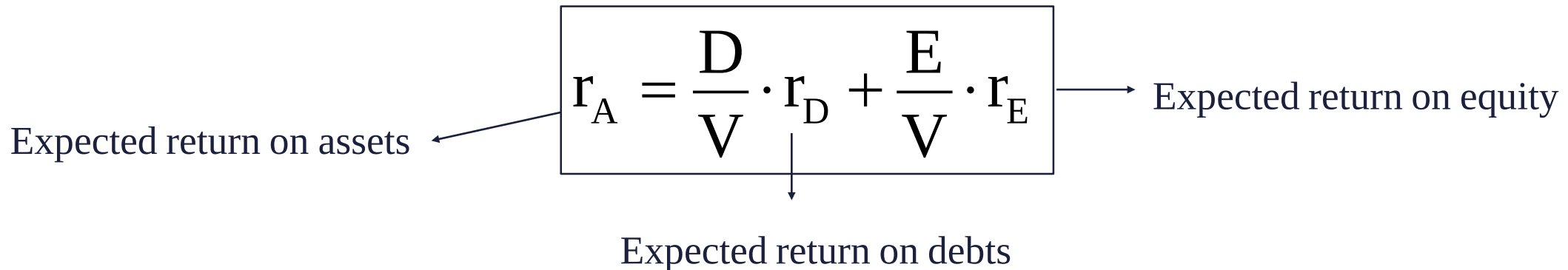
Solution: look for help on other markets

→ capital market → the expected rate of return can be measured there.

Expected return on assets

$$\text{Expected return of assets} = r_A = \frac{\text{expected operating income}}{\text{market value of all securities}}$$

$$= \frac{D \cdot r_D + E \cdot r_E}{V}$$


$$r_A = \frac{D}{V} \cdot r_D + \frac{E}{V} \cdot r_E$$

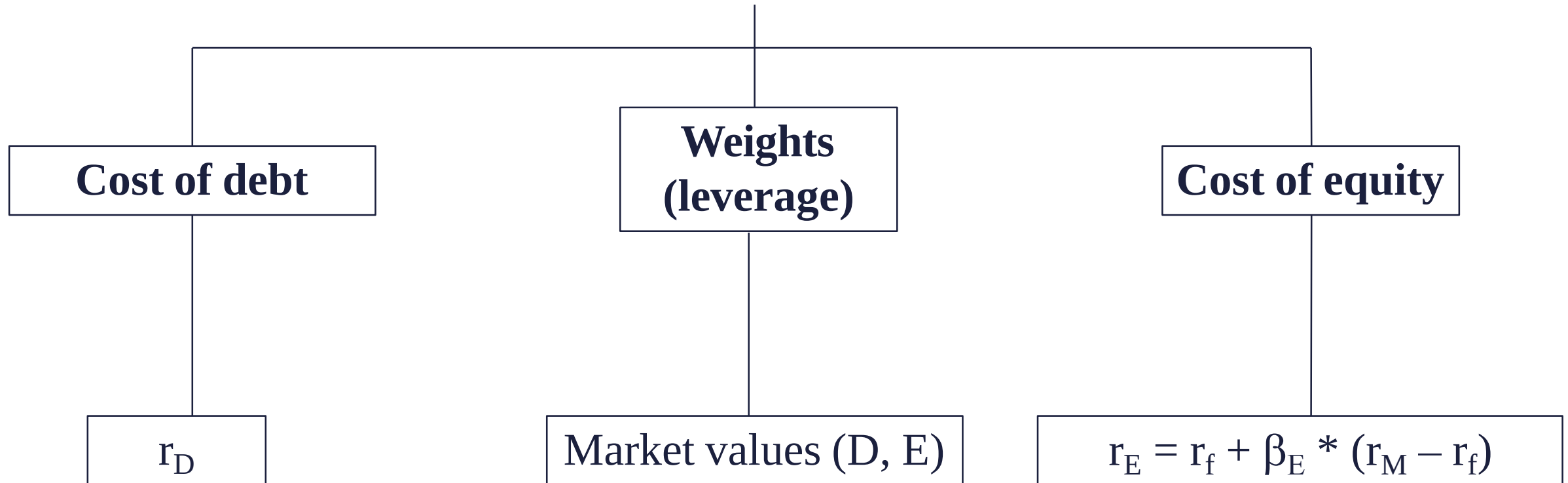
Expected return on assets

Expected return on equity

Expected return on debts

What is needed for the estimation?

$$r_A = \frac{D}{V} \cdot r_D + \frac{E}{V} \cdot r_E$$



Modigliani & Miller Theorems

Modigliani and Miller (MM) theorems have to follow conditions, referred to as **perfect capital markets**:

- **Investors and firms** can trade the **same set of securities** at **competitive market prices** equal to the present value of their future cash flows.
- There are **no taxes, transaction costs, or issuance costs** associated with security trading.
- A firm's **financing decisions do not change** the **cash flows generated by its investments**, nor do they reveal new information about them.

The original Modigliani-Miller article:

<https://www.jstor.org/stable/pdf/1809766.pdf?refreqid=excelsior%3Ad444cf2471fc5f618002ffbe69a96741>

Modigliani – Miller Theorems

Modigliani & Miller I. proposition $\rightarrow r_A = \frac{D}{V} \cdot r_D + \frac{E}{V} \cdot r_E$

In a perfect capital market, the total value of a firm is equal to the market value of the total cash flows generated by its assets and is not affected by its choice of capital structure.

Because: the total cash flow paid out to all of a firm's security holders is equal to the total cash flow generated by the firm's assets. $\rightarrow$ by the Law of One Price, the firm's securities and its assets must have the same total market value.

Homemade leverage: as long as investors can borrow or lend at the same interest rate as the firm, homemade leverage is a perfect substitute for the use of leverage by the firm $\rightarrow$ so why would someone give a higher value for a levered firm?

Proof – Homemade leverage

Unlevered firm: $V_U = E_U$

Buy 1% of unlevered firm's stock $\rightarrow$ 1% of profit

Dollar Investment	Dollar Return
$0.01 V_U$	$0.01 \times \text{Profits}$

Levered firm: $V_L = E_L + D_L$

Buy the same fraction both D and E

	Dollar Investment	Dollar Return
Debt	$0.01 D_L$	$0.01 \times \text{Interest}$
Equity	$0.01 E_L$	$0.01 \times (\text{Profits} - \text{Interest})$
Total	$0.01(D_L + E_L)$ $= 0.01 V_L$	$0.01 \times \text{Profits}$

Buy 1% of E_L

Dollar Investment	Dollar Return
$0.01 E_L$ $= 0.01(V_L - D_L)$	$0.01 \times (\text{Profits} - \text{Interest})$

Take 1% D_L and buy E_U from that

	Dollar Investment	Dollar Return
Borrowing	$-0.01 D_L$	$-0.01 \times \text{Interest}$
Equity	$0.01 V_U$	$0.01 \times \text{Profits}$
Total	$0.01(V_U - D_L)$	$0.01 \times (\text{Profits} - \text{Interest})$

Example 14.2 pp. 527. (Berk-DeMarzo, 2017)

Suppose there are two firms, each with date 1 cash flows of \$1400 or \$900. The firms are identical except for their capital structure. One firm is unlevered, and its equity has a market value of \$990. The other firm has borrowed \$500, and its equity has a market value of \$510. Does MM proposition I hold? What arbitrage opportunity is available using homemade leverage?

Assume that the risk free rate is 5%.

Example 14.2 pp. 527. – solution

(Berk-DeMarzo, 2017)

MM proposition: These firms hold identical assets, their total values should be the same.

- unlevered firm has a total market value of \$990
- the levered firm has a total market value of \$510 (equity) + \$500 (debt) = \$1010.
- **these prices violate MM proposition I.** → also the Law of One price → an arbitrage opportunity exists.

	Date 0	Date 1: Cash Flows	
	Cash Flow	Strong Economy	Weak Economy
Borrow	\$500	−\$525	−\$525
Buy unlevered equity	−\$990	\$1400	\$900
Sell levered equity	\$510	−\$875	−\$375
Total cash flow	\$20	\$0	\$0

Modigliani & Miller Proposition II.

The cost of capital of levered equity increases with the firm's market value debt-equity ratio.

$$r_E = r_A + \frac{D}{E} \cdot (r_A - r_D)$$

Modigliani & Miller Proposition II.

(Berk-DeMarzo, 2017)

Unlevered equity:

	Date 0	Date 1: Cash Flows		Date 1: Returns	
	Initial Value	Strong Economy	Weak Economy	Strong Economy	Weak Economy
Unlevered equity	\$1000	\$1400	\$900	40%	-10%



Expected return for the unlevered equity:
 $0.5 \times 40\% + 0.5 \times (-10\%) = 15\%$

Levered equity:

	Date 0	Date 1: Cash Flows	
	Initial Value	Strong Economy	Weak Economy
Debt	\$500	\$525	\$525
Levered equity	$E = ?$	\$875	\$375
Firm	\$1000	\$1400	\$900



Equity = Firm – Debt = 500
(because $V = D + E$)

why not the present value of the expected value... ??? Equity = $(0.5 \times 875 + 0.5 \times 375) / (1 + 15\%) = 543$

The problem is, that the risk is increasing for the levered equity! Why? ... next slide

Because...

Payments to debt holders are made before payments to equity holders



Risk is greater for the levered equity → return should be greater, too

How much is the return in the levered case? ... *next slide*

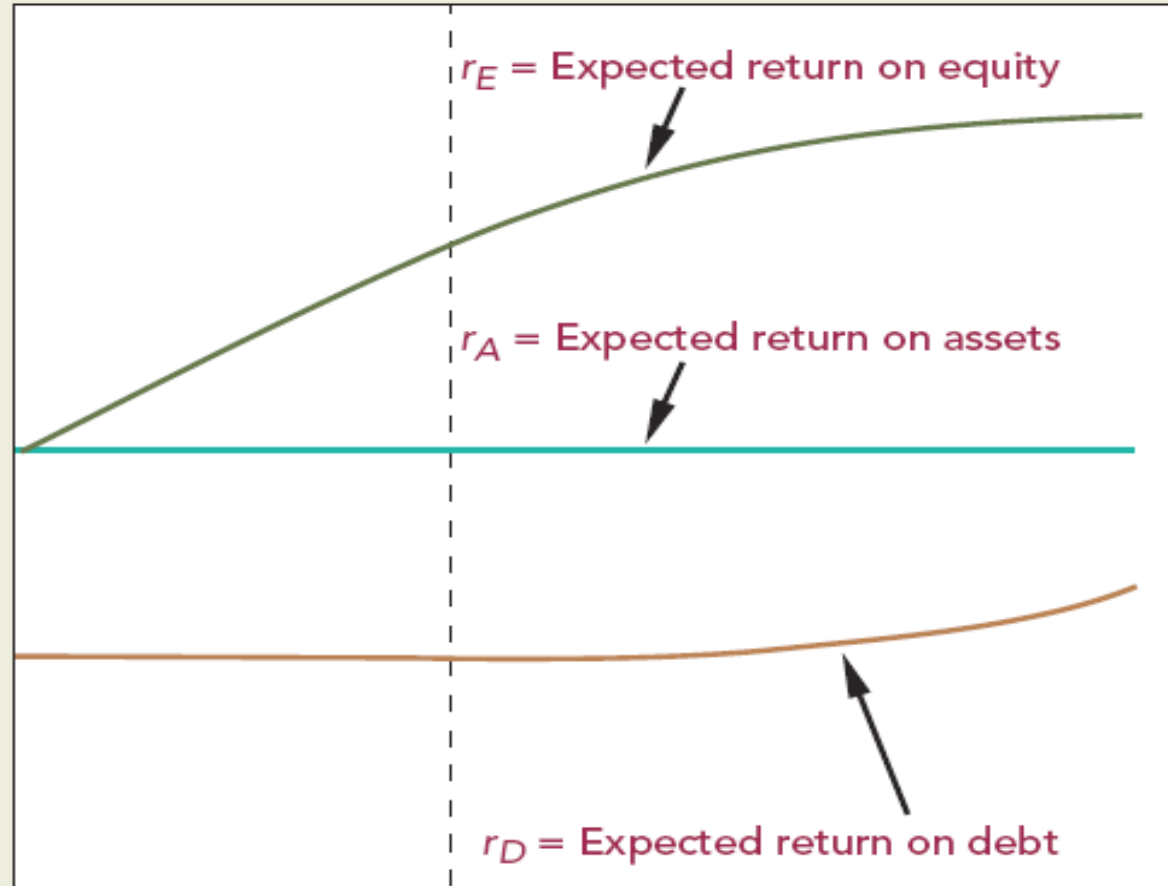
Return for the levered equity

	Date 0	Date 1: Cash Flows		Date 1: Returns		
	Initial Value	Strong Economy	Weak Economy	Strong Economy	Weak Economy	Expected Return
Debt	\$500	\$525	\$525	5%	5%	$r_D = 5\%$
Levered equity	\$500	\$875	\$375	75%	-25%	$r_E = r_L = 25\%$
Unlevered equity	\$1000	\$1400	\$900	40%	-10%	$r_A = r_U = 15\%$

If $D = 0 \rightarrow$ unlevered equity = assets $\rightarrow r_E = r_A = r_U$

Expected equity return is **increased to 25%** $\rightarrow$
leverage increases the risk of equity even when there is **no risk** that the **firm will default**

Rates of return



Risk-free debt $\longleftrightarrow$ Risky debt $\frac{D}{E} = \frac{\text{debt}}{\text{equity}}$

Leverage and Returns

Market Value Balance Sheet example

Asset Value 100	Debt (D) 40
	Equity (E) 60
Asset Value 100	Firm Value (V) 100

$$r_d = 7.5\%$$

$$r_e = 15\%$$

$$r_A = \left(r_D \times \frac{D}{D+E} \right) + \left(r_E \times \frac{E}{D+E} \right)$$

$$r_A = \left(.075 \times \frac{40}{100} \right) + \left(.15 \times \frac{60}{100} \right) = 12\%$$

Leverage and Returns

What happens if r_d increases?

Market Value Balance Sheet example

Asset Value 100	Debt (D) 40
	Equity (E) 60
Asset Value 100	Firm Value (V) 100

$$r_d = 7.875\%$$

$$r_a = 12\%$$

$$.12 = \left(.07875 \times \frac{40}{100} \right) + \left(r_e \times \frac{60}{100} \right)$$
$$r_e = 14.75\%$$

WACC – Weighted Average Cost of Capital

$$r_A = \left(r_D \times \frac{D}{D + E} \right) + \left(r_E \times \frac{E}{D + E} \right)$$

$$WACC = \left(r_D \times (1 - t) \times \frac{D}{D + E} \right) + \left(r_E \times \frac{E}{D + E} \right)$$

If $t=0$:

$$WACC = r_A$$

$$TS = 0$$

Example

$$D/V = 20\% \quad r_D = 10\% \quad r_E = 20\% \quad t = 50\%$$

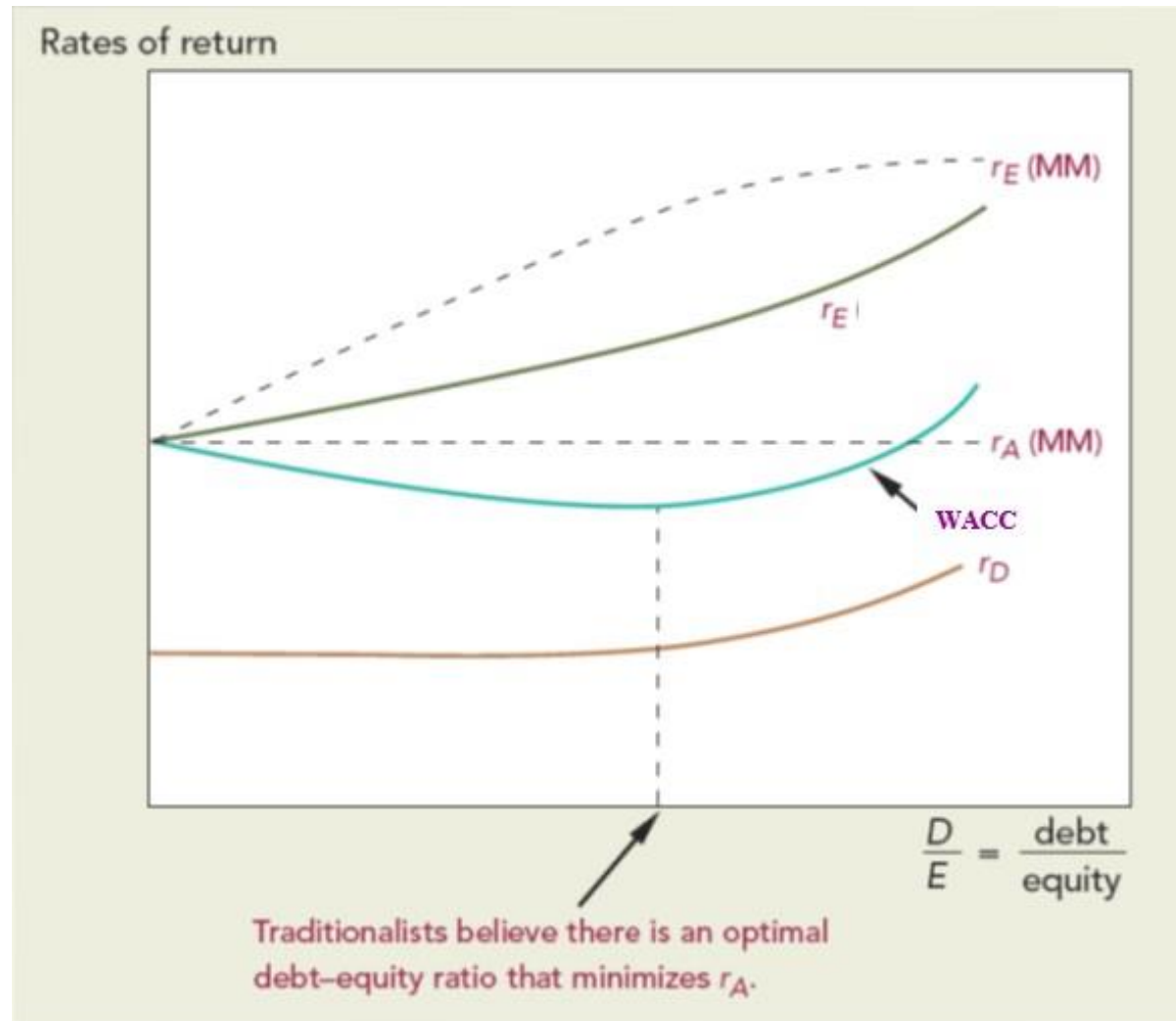
$$r_A = 20\% * 10\% + 80\% * 20\% = 18\%$$

$$WACC = 20\% * 10\% * (1 - 50\%) + 80\% * 20\% = 17\%$$

$$D/V = 40\% \quad r_D = 13\% \quad r_E = ?$$

$$r_E = (18\% - 40\% * 13\%) / 60\% = 21.33\%$$

$$WACC = 40\% * 13\% * (1 - 50\%) + 60\% * 21.33\% = 15.4\%$$



After Tax WACC

Example

The firm has a marginal tax rate of 35%. The cost of equity is 14.6% and the pretax cost of debt is 8%. Given the market value balance sheets, what is the tax adjusted WACC?

Balance Sheet (Market Value, millions)

Assets	125	50	Debt
		<u>75</u>	Equity
Total assets	125	125	Total liabilities

Example - continued

$$WACC = (1 - T_c) \left(\frac{D}{V} \times r_D \right) + \left(\frac{E}{V} \times r_E \right)$$

$$\begin{aligned} WACC &= (1 - .35) \left(\frac{50}{125} \times .08 \right) + \left(\frac{75}{125} \times .146 \right) \\ &= .1084 \\ &= 10.84\% \end{aligned}$$

Leverage and Risk

$$\beta_A = \left(\frac{D}{D + A} \times \beta_D \right) + \left(\frac{E}{D + E} \times \beta_E \right)$$

$$\beta_E = \beta_A + \frac{D}{E} (\beta_A - \beta_D)$$

Exercise 1.

The two units of Woodandog Works are Wood Factory and Dog Factory.

The market value of the Wood unit is 300 M €, and the industry's expected return is 20%.

The Dog Factory's market value is 100 M €, and the industry's expected return is 40%.

The company is funded by risk free credit by 40% of the company's market value, and its expected return is 10%.

a) Calculate the expected return of the company's assets and equity!

b) How does the expected return of the equity change if the company invest another 100 M € in the Wood unit financed by further equity rising? Explain the causes of the change in the expected rate of return!

$$\begin{aligned} a) \quad & V^{Wood} = 300; \quad V^{Dog} = 100; \quad w^{Wood} = 300/(300+100); \quad w^{Dog} = 100/(300+100) \\ & r(A)^{Wood} = 20\% \quad r(A)^{Dog} = 40\% \\ & r(A) = w^{Wood} * r(A)^{Wood} + w^{Dog} * r(A)^{Dog} = (300 * 20\% + 100 * 40\%) / 400 = 25\% \end{aligned}$$

$$D/V = 40\% \rightarrow D = D/V * V = 40\% * 400 = 160 \rightarrow E = 400 - 160 = 240$$

$$r(D) = 10\%$$

$$r(A) = r(D) * D/V + r(E) * E/V \rightarrow$$

$$r(E) = (r(A) - r(D) * D/V) / (E/V) = (25\% - 40\% * 10\%) / 60\% = 35\% \quad \text{OR}$$

$$r(E) = r(A) + (r(A) - r(D)) * D/E = 25\% + (25\% - 10\%) * (160/240) = 35\%$$

$$b) \quad r(A) = (400 * 20\% + 100 * 40\%) / 500 = 24\%$$

$$D = 160$$

$$r(E) = (24\% - (160/500) * 10\%) / (340/500) = 30.59\%$$

The return decreased because the ratio of the less risky business unit rose while the debt of the firm dropped.

Exercise 2

2. The production of our company, the Strawberry- Ice- Cream Works consists of two main products. The market value of Strawberry Cream Unit is 150 M €, and of the Cornet Unit is 300 M €. The company was financed by risk free credit by 20% of the market value. The data of the company's leading competitors are the following:

Competitor	Leverage in market value	Expected return of the credits	Expected return of the shares
Strawberry Cream Plc.	30%	12%	20%
Wafer Plc.	50%	15%	28%

The return of the risk free credit is 10%. The markets are perfect.

- Calculate the cost of capital for each unit!
- Determine the whole company's cost of the capital!
- Calculate the company's shares' expected rate of return!

Solution:

- a) *Strawberry Unit is being estimated from the data of the competitor, Strawberry Cream Plc. You have to calculate the $r(A)$ from the competitor's data. Based on MM I., the value of a company is independent from financing structure, so the competitor's $r(A)$ is the same for our company as well (it is working for the other unit also). Since the competitor and our unit is working in the same industry, the risk of the activity is the same, so the return should be the same as well. On equity and debt level you cannot assume the same, since the different capital structure can have an effect on the value of $r(E)$, $r(D)$, $\beta(D)$ or $\beta(E)$, because the risk is changing.*

$$\text{Strawberry Unit: } r(A)^{\text{Strawberry Unit}} = r(A)^{\text{Strawberry}} = r(D)^{\text{Strawberry}} * D/V^{\text{Strawberry}} + r(E)^{\text{Strawberry}} * E/V^{\text{Strawberry}} = 0.3 * 12\% + 0.7 * 20\% = 17.6\%$$

$$\text{Wafer Plc: } r(A)^{\text{Cornet}} = r(A)^{\text{Wafer}} = r(D)^{\text{Wafer}} * D/V^{\text{Wafer}} + r(E)^{\text{Wafer}} * E/V^{\text{Wafer}} = 0.5 * 15\% + 0.5 * 28\% = 21.5\%$$

$$b) \ r_A^{\text{Whole company}} = w^{\text{Strawberry unit}} * r(A)^{\text{Strawberry unit}} + w^{\text{Cornet}} * r(A)^{\text{Cornet}} = (150/450) * 17.6\% + (300/450) * 21.5\% = 20.2\%$$

$$c) \ r_E^{\text{Whole company}} = (20.2\% - 0.2 * 10\%) / 0.8 = 22.75\%$$

MM's Proposition III

- **With investment and production decisions given, the firm's financing decision is a matter of indifference for any of its security holders, since it does not affect its value.**

MM with Corporate Tax (1963)

Assumption: The payment of interest on debt is tax- exempt → debt provides a tax shield

$$V_L = V_U + t_c D$$

$$\begin{aligned} \text{PV(Interest tax shield)} &= \text{PV}(\text{tax} * \text{Future interest payments}) = \\ &= \text{tax} * \text{PV(Future interest payments)} = \text{tax} * D \end{aligned}$$

$$\blacksquare WACC = \frac{E}{E+D} r_E + \frac{D}{D+E} r_D (1 - t_c)$$

$$\blacksquare WACC = \frac{E}{E+D} r_E + \frac{D}{D+E} r_D - \frac{D}{E+D} r_D t_c$$

MM with Corporate Tax (1963)

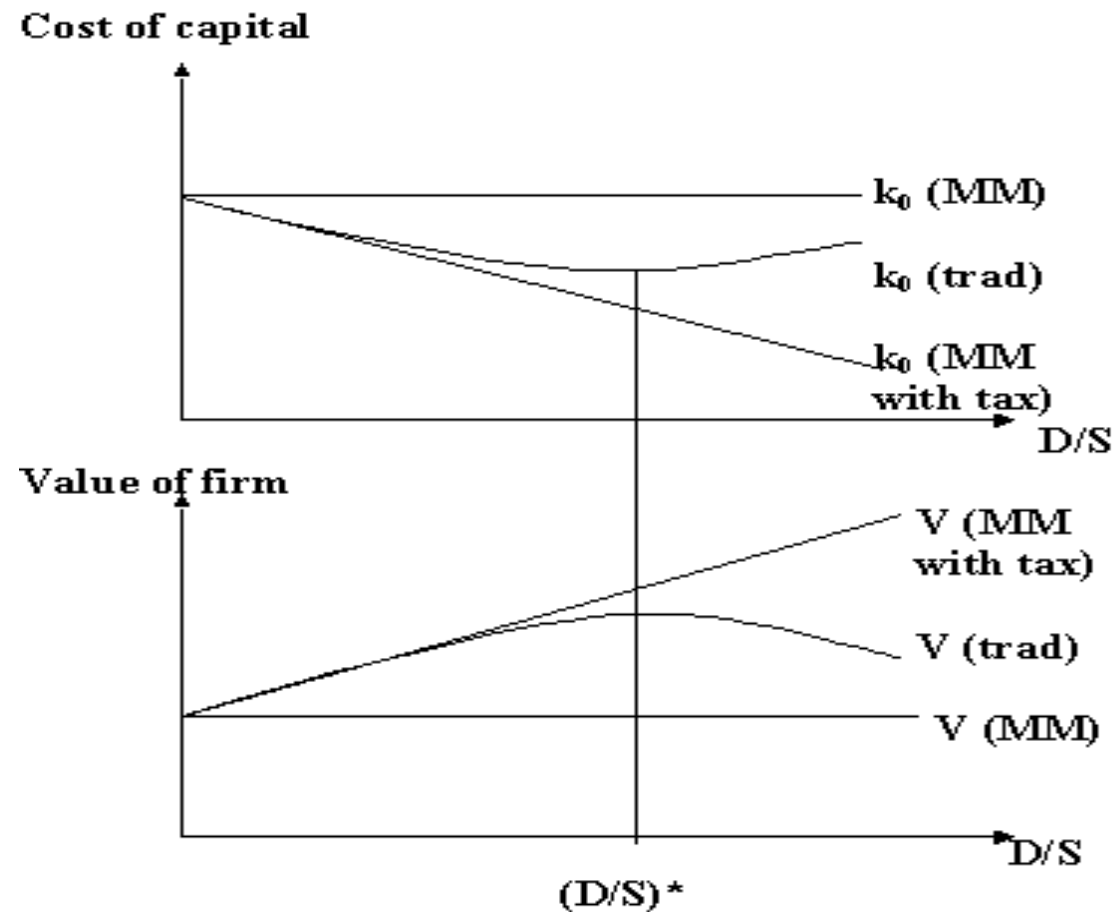
Assumption: The payment of interest on debt is tax- exempt → debt provides a tax shield

$$V_L = V_U + t_c D$$

$$k_{0,L} = k_0 - t_c(k_0 - i) D/V_L = \text{after-tax } k_0$$

$$k_{e,L} = k_0 + (1 - t_c)(k_0 - i) D/S_L = \text{after-tax } k_e$$

Comparing theoretical predictions:



Capital Structure with Corporate & Personal Taxes (Miller 1997)

$$V_L = V_U + D \left[1 - \frac{(1 - t_c)(1 - t_{ps})}{(1 - t_{pb})} \right]$$

where:

t_c = corporate tax rate

t_{ps} = tax rate on income from shares

t_{pb} = tax rate on income from bonds

Comparison of the predictions

If $t_c = t_{ps} = t_{pb} = 0 \Rightarrow V_L = V_U$ (MM 1958)

If $t_{ps} = t_{pb} \Rightarrow V_L = V_U + t_c D$ (MM 1963)

If $t_{ps} < t_{pb} \Rightarrow \text{benefits from_leverage} < t_c D$



**Thank you
for your attention!**